

MEWNI ALAHAKOON

📍 Colombo, Sri Lanka 📞 +94 78 621 3499 ✉ mewnialahakoon@gmail.com 🌐 mewni.github.io
🐙 github.com/mewni 📄 mewni-a-0a7a48192 🆔 0009-0006-4055-3090 🎓 Google Scholar

EDUCATION

- Master of Information Security (MIS)** 2023 – 2025
University of Colombo School of Computing (UCSC), University of Colombo Colombo, Sri Lanka
- Thesis: *Smart Card Vulnerabilities in Sri Lanka: A Security Assessment Across Service Providers*. Published in Springer CCIS [1].
 - Supervisor: Dr. Asanka P. Sayakkara.
- BSc (Hons) Computer Security**, Upper Second-Class Honours (2:1) 2019 – 2022
University of Plymouth (delivered at NSBM Green University, Sri Lanka) United Kingdom
- Thesis: *A Machine Learning Approach to Detect Zero-Day Ransomware Attacks*.
 - Supervisor: Mr. Chamindra Attanayake.

RESEARCH INTERESTS

Contactless (NFC/EMV) payment security, EMV card security testing, relay attacks, long-range radio interception, online payment and payment-API security, payment-ecosystem security, peripheral and operating-system security (USB, USB-C, Thunderbolt), embedded and IoT security, experimental security evaluation with SDR, threat modelling.

PUBLICATIONS

Peer-reviewed

- [1] **M. Alahakoon**, A. P. Sayakkara, B. Weerasinghe, and A. Wickramasekara. “Smart Card Vulnerabilities in Sri Lanka: A Security Assessment Across Service Providers.” In *Proceedings of the International Conference on Advances in ICT for Emerging Regions (ICTer 2025)*, Communications in Computer and Information Science (CCIS), Springer, 2026. doi:10.1007/978-981-92-0137-2_31.
- [2] **M. Alahakoon**. “Weaponizing the Unknown: Zero-Day Exploits as Tools of Cyberwarfare and Their Impact on Global Security.” *BCIS Emerging Scholars Symposium (ESS 2025)*, Bandaranaike Centre for International Studies, Colombo, Sri Lanka. [**Forthcoming**; presented at the symposium.]

Work in progress

- [3] **M. Alahakoon** and A. P. Sayakkara. Systematic review of USB security: attacks, defences and open challenges.
Literature review in progress; survey article planned.

RESEARCH EXPERIENCE

- Independent Researcher**, USB and Peripheral-Interface Security 2026 – Present
In collaboration with Dr. Asanka P. Sayakkara, UCSC, University of Colombo Colombo, Sri Lanka
- Conducting an ongoing systematic literature review of USB security [3], covering USB4, USB-C Power Delivery, Thunderbolt, and malicious cable implants.
 - Developing an attack-layer × attacker-capability taxonomy that spans the physical to the application layer, and surveying host-side defences such as device authorisation, firmware signing and kernel DMA protection.
 - Mapping open problems where attack classes remain unstudied or proposed defences lack independent evaluation.

Postgraduate Researcher, Master's Thesis

2024 – 2025

UCSC, University of Colombo · Supervisor: Dr. Asanka P. Sayakkara

Colombo, Sri Lanka

- Led the first systematic, cross-provider security assessment of NFC payment cards issued in Sri Lanka, covering cards from nine banks under a defined proximity-attacker threat model [1].
- Designed an original quantitative security-scoring model combining read success, read latency, distance resistance and wallet-shielding resistance across 11 experimental scenarios.
- Ran data-extraction (Flipper Zero), jamming (nine cards) and replay (five cards) experiments at 13.56 MHz with HackRF One and GNU Radio; jamming was defeated by EMV anti-collision and signal modulation, while replay was limited by SDR frequency drift.
- Showed that the principal risk lay in the wider payment ecosystem rather than in card cryptography: card data exposed over NFC could be misused where online merchants apply incomplete validation.

Postgraduate Researcher, Zero-Day Exploits in Cyberwarfare

2025

UCSC, University of Colombo · Sole-authored study [2]

Colombo, Sri Lanka

- Conducted a qualitative comparative case study of Stuxnet, WannaCry and Pegasus, using thematic coding of state motives, offence–defence imbalance and security impact.
- Showed how stockpiling and the exploit market create an offence–defence asymmetry; assessed governance gaps against the U.S. Vulnerabilities Equities Process and the Tallinn Manual 2.0.

Undergraduate Researcher, BSc Thesis

2021 – 2022

University of Plymouth (delivered at NSBM Green University)

Homagama, Sri Lanka

Supervisor: Mr. Chamindra Attanayake

- Designed a real-time, multi-stage detection architecture for zero-day ransomware on Windows, combining pre-execution static analysis of downloaded executables with behavioural monitoring of file, directory and registry activity.
- Implemented the static-analysis stage as a two-model pipeline: a random-forest classifier over PE-file features and a convolutional neural network trained on 9,639 malware images from 25 families (Microsoft malware dataset).
- Deployed both models as a FastAPI service (Python, scikit-learn, TensorFlow/Keras).

OTHER RESEARCH ACTIVITIES

Student Volunteer Researcher, Conex Research

2020 – 2021

International, UK-based student research organisation

Remote

- Contributed to space-mission proposals through background research, Python scripting, and mission analysis in NASA's GMAT.

TECHNICAL SKILLS

Hardware & Radio	Flipper Zero, HackRF One, GNU Radio, GQRX
Protocols	NFC/RFID, EMV, ISO/IEC 14443, USB, USB-C
Security Tools	Wireshark, Burp Suite, Nmap, Metasploit, Nessus
Digital Forensics	Autopsy, FTK Imager
Programming	Python, C, C#, JavaScript, PHP, SQL
Machine Learning	scikit-learn, TensorFlow/Keras
Operating Systems	Linux (Kali, Ubuntu, Arch), Windows

SERVICE & MEMBERSHIPS

Team Lead, Software Freedom Day 2020, NSBM Green University

2020

Volunteer and Senior Member, FOSS Community, NSBM Green University

2019 – 2022

Member, Mozilla Campus Club, NSBM Green University

2019 – 2022

Student Member, IEEE (Computer Society; Women in Engineering)

2021 – 2022